Name: Devansh Singh
PRN: 25070122072
Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 9

TITLE : Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden and a final class cannot be inherited.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp9/Main.java
Name : Devansh
ID : 101
College : Symbiosis Institute of Technology
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create a Bank Account program where account number is final and cannot be changed.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp9/Exercise/Ex1.java
Account Number : 123456
Account Holder : Devansh
Balance        : 50000.0
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp9/Exercise/Ex2.java
ISBN   : 978-93-12345-67-8
Title  : Java Programming
Author : James Gosling
Price  : 599.0
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ 
```

Conclusion

This assignment helped in understanding the concepts of inheritance and interfaces in Java through real-world examples. It demonstrated how inheritance enables code reusability by allowing child classes to inherit the properties and methods of a parent class, while interfaces provide abstraction by defining common behaviors that can be implemented by different classes. The implementation improved my understanding of object-oriented programming principles such as reusability, abstraction, and polymorphism. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, e-commerce systems, banking systems, and other real-world Java applications that require modular, flexible, and maintainable code.